

---

---

REVIEW

---

---

## T-Regulatory Cells as Part of Strategy of Immune Evasion by Pathogens

F. Yu. Garib<sup>1,2\*</sup> and A. P. Rizopulu<sup>3</sup>

<sup>1</sup>Lomonosov Moscow State University, Biological Faculty, 119991 Moscow, Russia

<sup>2</sup>Russian Medical Academy of Postgraduate Education, Department of Immunology,  
123995 Moscow, Russia; E-mail: fgarib@yandex.ru

<sup>3</sup>Committee on Science and High Technologies, State Duma of the Federal Assembly  
of Russian Federation, 103265 Moscow, Russia; E-mail: annarizopulu@inbox.ru

Received November 20, 2014

**Abstract**—Under physiological conditions, regulatory processes can suppress the immune response after elimination of a pathogen and restore homeostasis through the destruction and suppression of obsolete effector cells of the immune system. The main players in this process are T-regulatory cells (Tregs) and immature dendritic cells, which suppress the immune response by their own products and/or by inducing synthesis of immunosuppressive interleukins IL-10, IL-35, and transforming growth factor (TGF- $\beta$ ) by other cells. This mechanism is also used by widespread “successful” pathogens that are capable of chronically persisting in the human body — herpes virus, hepatitis viruses, human immunodeficiency virus, *Mycobacterium tuberculosis*, *Helicobacter pylori*, and others. During coevolution of microbial pathogens and the host immune system, the pathogens developed sophisticated strategies for evading the host defense, so-called immune evasion. In particular, molecular structures of pathogens during the interaction with dendritic cells via activating and inhibitory receptors can change intracellular signal transduction, resulting in block of maturation of dendritic cells. Immature dendritic cells become tolerogenic and cause differentiation of Tregs from the conventional T-cell CD4<sup>+</sup>. Microbial molecules can also react directly with Tregs through innate immune receptors. Costimulation of Toll-like receptor 5 (TLR5) by flagellin increases the expression of the transcription factor Foxp3, which increases the suppressive activity of Treg cells. From all evasion mechanisms, the induction of immunosuppression by Treg through IL-10, IL-35, and TGF- $\beta$  appears most effective. This results in the suppression of inflammation and of adaptive immune responses against pathogens, optimizing the conditions for the survival of bacteria and viruses.

DOI: 10.1134/S0006297915080015

**Key words:** Treg, CD4<sup>+</sup>CD25<sup>high</sup>Foxp3<sup>+</sup>, antigen-presenting cells, IL-10, CTLA-4, immunosuppression, immune evasion

### GENERAL MECHANISMS OF IMMUNE EVASION BY PATHOGENS

In most cases, infectious processes end up with elimination of pathogens, which are successfully dealt with by the immune system. However, the efficacy of immune response is not always sufficient to eradicate a

pathogen that creates conditions when the disease acquires a chronic course. Persistent infectious agents such as herpesvirus, hepatitis viruses, human immunodeficiency virus, *Mycobacterium tuberculosis*, *Helicobacter*, prions, fungi, as well as parasites are able to survive in the host's body without causing overt disease symptoms. Billions of people may be carriers of patho-

---

**Abbreviations:** B7, a costimulatory molecule expressed on the surface of antigen-presenting cells (APCs); CD4<sup>+</sup>, a marker expressed by helper and regulatory T-cells; CD4<sup>+</sup>CD25<sup>high</sup>Foxp3<sup>+</sup>, markers expressed by regulatory T-cells; CD4<sup>+</sup>Th2, type 2 helper CD4<sup>+</sup> T-cells; CD4<sup>+</sup>Treg or CD8<sup>+</sup>Treg, regulatory T-cells; CD8<sup>+</sup>, a marker expressed by cytotoxic T-cells; CTLA-4, an inhibitory molecule expressed by T-cells; FHA, bacterial filamentous hemagglutinin adhesin; HIV, human immunodeficiency virus; IL-10 and IL-35, interleukins; LFA-3 and ICAM-1, cell adhesion molecules; MHC I and II, molecules of the major histocompatibility complex; PD1, an inhibitory molecule expressed on dendritic cells (programmed cell death protein 1); STAT, intracellular signaling pathway (signal transducer and activator of transcription); TAP, transporter associated with antigen processing expressed in APCs; TCR, T-cell receptor; TGF- $\beta$ , transforming growth factor- $\beta$ ; Th1, type 1 helper CD4<sup>+</sup> T-cells; TLR, Toll-like receptor.

\* To whom correspondence should be addressed.

gens causing long-lasting latent infections that have become widespread in the human population. Undoubted “successfulness” of pathogens is accounted for by various and numerous ways they use to avoid immune reactions, generally described as “immune evasion”, which has been extensively studied in recent years [1, 2].

All pathogenic microorganisms go through evolutionary selection based on their ability for reproduction and spread of their own genome upon parasitization in the host's body. This has resulted in gradual upgrade of immune reactions as well, that in turn faces resistance due to accelerated evolution of pathogens, and provides new impetus for further improvement of immune mechanisms. Based on this viewpoint, it is clear that coevolution preserves vital and reproductive functions of both the parasite and the host (the latter an obligatory source for the life of the parasite).

Previously unknown invasion factors allowing pathogens to not only escape from immune recognition, but also even to manage immune processes have been found using molecular genetic analysis. The interaction of pathogens with innate immunity occurs at early stages of infection, thus being a crucial period for survival of pathogens. In particular, to escape from recognition by innate immune receptors, pathogenic bacteria apply the following mechanisms: encapsulation, by decorating the bacterial surface with host proteins, by varying molecular structures of the cellular wall, etc. Moreover, bacteria can survive and multiply inside phagocytes after engulfment by blocking phagosome maturation and impacting intracellular signaling pathways [3-5].

To dampen protective inflammatory responses, pathogens inhibit production of host proinflammatory cytokines and secrete their own proteases that degrade complement components, interleukins, interferons, chemokines, antimicrobial peptides, etc. [2]. By using a “molecular syringe”, pathogens regulate functions of immune cells by injecting certain effector molecules encoded within special “pathogenicity islands” belonging to type III and IV secretion systems. To do this, bacteria create new genetic structures built from the external material obtained from the environment (most often via bacteriophages) [6].

One of the evasion factors used by bacterial pathogens is a unique system of “bacterial community interactions” called “quorum sensing”, when each bacterium produces some signaling molecules (QSM, quorum sensing signaling molecules) and recognizes them by using specialized sensors. By doing this, each bacterium can determine the overall “density” of the same kind of cells, so that an “organized” invasive process starts only in the presence of a bacterial “quorum”. QSM is determined by the *luxS* gene that also controls secretion of virulence effector molecules suppressing immune reactions [7, 8].

The evasion program used by viral pathogens (cytomegalovirus, vaccinia, Epstein–Barr virus, rabbit myxoma virus, etc.) is featured as follows: virally encoded soluble homologs of cytokines, interferons, and chemokines and their receptors produced by the infected host cells; expression of adhesion molecules such as LFA-3 and ICAM-1 downregulated on interacting host cells collectively resulting in neutralization of the host defense mechanisms, lowered magnitude of inflammatory process, and abandoned interferon-dependent antiviral effects [9]. Viruses, being obligate intracellular parasites, induce expression of inhibitory receptors on natural killer cells (NK cells), which hinders their ability to recognize and lyse infected host cells. Tactics deployed by viruses in dendritic cells and macrophages during the first step of adaptive immune response to inhibit all stages of processing and presentation of viral antigens to cytotoxic CD8<sup>+</sup> T-cells is of special interest. In particular, herpes simplex virus, cytomegalovirus, etc. affect proteasome functioning followed by inhibiting TAP complex as well as by blocking binding of viral epitopes to MHC class-I molecules by destroying MHC class-I molecules and/or inhibiting their production inside the endoplasmic reticulum [10, 11].

It is known that immune reactions are regulated by a complex multifaceted system. Molecular structures of pathogens and antigens as well as “danger signals” released by stressed and damaged tissues induce immune response. After recognizing pathogen molecular patterns, innate immune reactions are triggered, which are usually sufficient to eliminate infectious pathogens. Then, a more specific humoral and cellular adaptive immune response accompanied by development of memory lymphocytes develops if necessary. It must be emphasized that immune response is characterized by redundant defense mechanisms. It often results in abundance of effector functions and represents a real threat to damage tissues and organs, which is manifested as hyperinflammatory and autoimmune reactions. Therefore, mere survival of the body during infection requires development of optimal immune response not only aimed at eradication of invasive pathogens, but also restricting damage to their own tissues, especially under extremely strong immune response [10]. This is why suppression of immune response represents an important regulatory mechanism. In most cases, immunosuppression takes place after elimination of pathogens, which is necessary to complete the immune process and represents an important homeostatic factor preventing development of immune pathology [12, 13].

Interestingly, pathogens may co-opt the host's regulatory mechanisms and enhance immunosuppression, starting at early stages of infection, thus providing pathogen survival and latent persistence during an entire human life and a high consequent probability to infect the progeny of the host.

## ORIGIN AND MOLECULAR MARKERS OF REGULATORY T-CELLS

The appropriate pattern of immune response is determined by the type of pathogen as well as its location (intracellular or extracellular). The fight against intracellular pathogens (viruses, intracellular bacteria) involves cellular immune response including type 1 helper CD4<sup>+</sup> T-cells (Th1), cytotoxic CD8<sup>+</sup> T-cells, and activated macrophages. Antibody-dependent immune response controlled by type 2 and 17 helper CD4<sup>+</sup> T-cells (Th2, Th17) is necessary to fight intracellular pathogens. At the same time, immunity against various types of pathogens is based on activation of Th17-mediated inflammatory processes, which, in turn, involve Th1- and Th2-cells. An important role in regulating immune processes is played by antigen-presenting cells including dendritic cells, which determine the type of the forthcoming helper T-cell response (Th1, Th2, Th17, etc.) appropriate for a particular pathogen. In case an immune reaction might be more dangerous than pathogen-associated effects, antigen-specific tolerance is developed via differentiation of regulatory T-cells.

Different types of regulatory T-cells (Tregs) have been described based on their origin and mode of action, which include two major populations: natural (nTreg) and induced (iTreg) Tregs.

Natural Tregs (nTreg, or thymus-derived Tregs, tTreg) are regulatory CD4<sup>+</sup> T-cells that differentiate directly inside the thymus. The classical understanding about the regulatory (suppressive) function of regulatory T-cells emerged while examining mechanisms used to maintain immune tolerance towards components of the host's body, i.e. self-antigens. It is known that random rearrangements within the germline gene segments V(D)J encoding  $\alpha$ - and  $\beta$ -T-cell receptor chains (TCR) occur during differentiation of thymocytes. T-cells developing in the thymus bear TCRs recognizing not only foreign antigens, but self-antigens as well. Autoreactive T-cells are depleted during positive and negative selection via anergy and apoptosis, thus deleting harmful clones. However, some self-reactive CD4<sup>+</sup> T-cells survive owing to production of antiapoptotic Bcl-2 protein, and they acquire a regulatory function. Such cells possess the stable committed phenotype CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> [10].

Induced Tregs (iTreg) develop from T-cells in peripheral lymphoid organs under the influence of various factors, including contacts with dendritic cells, macrophages, as well as IL-10 and TGF- $\beta$ , etc.

Both Treg populations (nTregs and iTregs) have high-affinity receptors to self-antigens that give them priority (compared to self-reactive helper and cytotoxic T-cells) upon interactions with antigen-presenting macrophages. After that, activated Tregs produce IL-10, IL-35, and TGF- $\beta$  that act to suppress autoimmune reactions [10].

In the case of genetic deficiency of targeting CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> cells, fatal autoimmune inflammation is developed [10-15], which shows the crucial role for Tregs in preventing autoimmune processes.

Along with that, iTregs bearing TCRs specific to foreign antigens develop during an immune process against pathogens. Such iTregs are also peripheral CD4<sup>+</sup> or CD8<sup>+</sup> T-cells, which acquire regulatory functions during immune response by starting expression of Foxp3 transcription factor and/or production of suppressive cytokines such as TGF- $\beta$ , IL-10, or IL-35. Importantly, iTregs can be converted from effector antigen-specific Foxp3<sup>-</sup> into Foxp3<sup>+</sup> T-cells, resulting in increased range of specific iTregs against exogenous antigens, thus providing effective control after immune reactions targeting various pathogens. In contrast to thymus-derived Tregs (nTregs), iTregs are not programmed in a permanent manner, and under certain circumstances, they can return to effector phenotype while losing expression of CD25 and Foxp3 and transforming towards Th1, Th2, and CD8<sup>+</sup> T-cells. Thus, CD4<sup>+</sup> and CD25<sup>+</sup> (IL-2R  $\alpha$ -chain) and/or Foxp3 are expressed by the dominant Treg population. Upon activation, they produce cytokines including TGF- $\beta$ , IL-10, or IL-35 and depend on IL-2 to grow and survive. Foxp3<sup>+</sup> is a transcription factor that regulates expression of the genes responsible both for their differentiation program and expression of molecules involved in suppression of immune response [13, 14].

Usually, CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> cells bear CTLA-4 (cytotoxic T-lymphocyte-associated protein 4) receptor possessing inhibitory activity on target cells as well as GITR (glucocorticoid-induced tumor necrosis factor receptor) able to activate both regulatory and nonregulatory T-cells. GARP (glycoprotein-A repetitions predominant) appear on activated human Tregs, which influences the development of suppressive properties typical to Tregs [15].

It should be noted that a novel subpopulation of regulatory T-cells has been recently found in mice and humans called Tr35 that does not express Foxp3 and develops upon conditioning with IL-35. In turn, they produce the suppressive cytokine IL-35, but neither IL-10 nor TGF- $\beta$ . In recent years, IL-35 has been considered as a powerful and stable factor of immune suppression. It is noted that *in vivo* Tr35 cells develop upon the influence of proinflammatory factors in the gut wall during infection with the helminth *Trichuris muris*, as well as in the microenvironment of malignant tumors such as melanoma and MC38 colorectal adenocarcinoma. It was assumed that IL-35 is a key mediator inducing infectious and tumor-derived tolerance that may have important significance for its therapeutic application in the future [16].

Another (quite effective) subpopulation of regulatory T-cells known as Tr1 produces IL-10, which mainly acts as an immunosuppressive cytokine on various immune cells. At the same time, IL-10, by acting on den-

dritic cells, suppresses the Th1-skewed differentiation program, thereby promoting cellular immune response and tilting it towards Th2-differentiation, i.e. enhancing humoral immune response. During protozoal infection, IL-10 may be produced not only by Tr1, but effector Th1- and Th2-cells as well [17], thus the use of IL-10 production as the only marker for determining Tr1-cells may not always be consistent.

Despite the fact that Tregs belong to CD4<sup>+</sup> T-cells, CD8<sup>+</sup> T-cells also can express Foxp3 and produce suppressive cytokines [18, 19].

Thus, a pool of regulatory T-lymphocytes contains quite heterogeneous cells including several subpopulations able to suppress immune reactions. Suppressive potential of regulatory T-cells is primarily supposed to act for maintaining self-tolerance, i.e. preventing autoimmune reactions and excessive inflammation, as well as restoring homeostasis after immune response. Until now, no specific surface markers for regulatory T-cells have been determined, whereas those currently used in practice to identify such cells *in vitro* and *in vivo* provide only to a certain degree of confidence. Based on this, the search for specific surface markers for Tregs and their subpopulations represents a problem to be solved to reveal novel mechanisms of immunoregulation including suppression, as well as the opportunity to use them for immunotherapy of autoimmune and allergic diseases.

Perhaps, understanding mechanisms underlying induction and maintenance of regulatory T-cells viewed from the perspective of pathogen evasion will help to develop effective modalities acting on these cells to treat patients with various diseases characterized by hypo- or hypersuppression.

#### MECHANISMS OF SUPPRESSIVE ACTIVITY OF REGULATORY T-CELLS

Tregs can suppress immune response at all stages starting from early innate immune response as well as during induction of T-cell activation, proliferation, and differentiation in lymphoid organs, and during the effector phase in non-lymphoid tissues.

Suppressive activity of CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> T-cells targets various types of cells, primarily dendritic cells, as well as macrophages, osteoblasts, mast cells, NK cells, CD4<sup>+</sup> and CD8<sup>+</sup> T-cells, B-cells, and natural killer T-cells (NKT-cells) [20]. Several mechanisms of suppressive activity have been described upon direct cell–cell contacts between Tregs and their target cells. In particular, Tregs were shown to produce cytokines IL-10, TGF- $\beta$ , and IL-35 that inhibit vital processes in effector T-lymphocytes and other immune cells [21, 22].

Simultaneously, it was found that in *in vivo* and *in vitro* experiments activity of effector T-cells was directly inhibited via membrane-bound TGF- $\beta$  on Tregs [23].

It is known that IL-2 is the major growth factor for various T-cell subsets. Tregs possess an enhanced ability to capture IL-2 due to upregulated surface expression of IL-2R  $\alpha$ -chain (CD25). Relatively high affinity of the IL-2R promotes preferable uptake of IL-2 by Tregs with its depletion in the surroundings, whereas shortage of IL-2 inhibits proliferation of effector T-cells and results in cell apoptosis [20].

A perforin-dependent cytotoxicity mediated by Tregs is one of the mechanisms underlying their direct suppressive action that targets direct participants of the immune response such as activated T-cells, monocytes, dendritic cells, and NK-cells [24].

Galectin-1 belonging to lectin protein family located on the surface of activated Tregs is able to bind to the relevant ligands on effector T-cells, which inhibits their proliferation, lowering production of IL-2 and IFN $\gamma$  and causing apoptosis [25].

More often, a mechanism of indirect immune suppression is accomplished via dendritic cells. By using different molecules, Tregs act on dendritic cells, which in turn suppress functions of other cells. In particular, CTLA-4 on Tregs binds to costimulatory molecules CD80 and CD86 on dendritic cells by blocking them or removing them from the cell surface via endocytosis, thus lowering their availability for naive T-cells and hindering costimulation during antigen presentation followed by development of anergy and apoptosis in antigen-specific helper and cytotoxic T-cells [20]. Obviously, such interactions limit the ability of dendritic cells to activate naive T-cells and lowers efficacy of specific immune response. Moreover, by acting via CTLA-4, Tregs stimulate antigen-presenting cells to produce indoleamine-2,3-dioxygenase (IDO), which degrades tryptophan. Lack of this essential amino acid in T-cell zones inside the lymph node inhibits activation of effector T-cells and leads to their apoptosis [20].

Also, another mechanism of Treg-dependent suppressive activity is mediated via cAMP and adenosine. In particular, it is considered that Tregs can act on other cells by changing cAMP level inside target cells: high cAMP level is associated with inhibited cell proliferation and differentiation. Moreover, it is accompanied by downregulated expression of key cytokines necessary for development of immune response, particularly IL-2 and IFN $\gamma$ . Tregs can elevate cAMP level even by directly delivering it into activated target cells via a gap junction [26]. On the other hand, Tregs can produce adenosine by using ectonucleotidases CD73 and CD39 expressed on their surface, so that extracellular adenosine interacts with several cellular receptors on target cells and exhibit a powerful antiinflammatory effect. G-protein coupled adenosine receptor A2A and A2B (A2AR and A2BR) become activated in the presence of extracellular adenosine and transmit a signal resulting in upregulation of cAMP. In addition, signals from activated A2AR (having the highest

affinity to adenosine) inhibit the TLR-triggered proinflammatory signaling cascade [26-28]. It is interesting that for evasion *S. aureus* expresses a cell wall-anchored adenosine synthase A (AdsA) that converts adenosine monophosphate to adenosine. These bacteria use immunosuppressive properties of adenosine it produces to inactivate phagocytosis, thus escaping from phagocytic clearance. A mutant *S. aureus* lacking AdsA expression has lower survival, which can be compensated by adding exogenous adenosine. Moreover, ten more Gram-positive bacterial species (e.g. *Clostridium perfringens* and *Listeria monocytogenes*) expressing AdsA homologs were identified. A study with *B. anthracis* demonstrated that it also uses adenosine to escape from phagocytosis, indicating that the AdsA-mediated evasion mechanism might be a quite common feature among bacteria [29, 30]. Given that signaling from A2AR inhibits Th1 and Th17 cells and contributes to development of adaptive Tregs (aka iTregs), AdsA-positive pathogens might perhaps manipulate T-cell immunity and IL-17-dependent inflammation, accompanied by the consequent recruitment of neutrophils.

Other surface molecules on T-cells such as LAG-3 and NRP-1 are also important for suppressive function of Tregs mediated via dendritic cells. LAG-3 (CD223) is a CD4 homolog that binds with high affinity to MHC class II molecules presenting antigen epitopes on dendritic cells. Consequently, it results in an intracellular inhibitory signaling pathway that delays maturation of dendritic cells, followed by lowering their antigen-presenting and costimulatory activities. At the same time, by acting via LAG-3, Tregs can directly suppress functions of activated T-cells expressing MHC class II molecules on their surface [31].

Neuropilin-1 (NRP-1) expressed on Tregs contributes to a prolonged interaction between Treg and immature dendritic cells, which creates conditions for development of a new generation of Tregs if the dendritic cells were simultaneously loaded with a low tolerogenic dose of antigen [32].

Thus, the ultimate targets for suppressive activity of Tregs are the cells mediating antigen-specific and inflammatory reactions of innate and adaptive immunity. Thus, Tregs can provide their suppressive activity directly and indirectly via dendritic cells. Direct Treg-dependent suppression is mainly mediated via inhibition of the cell cycle through numerous pathways: by producing suppressive cytokines such as IL-10, TGF- $\beta$ , IL-35; by expressing membrane-bound TGF- $\beta$ , galectin-1, CTLA-4, LAG-3, NRP-1; by delivering cAMP inside the target cells and producing extracellular adenosine; by inducing apoptosis via depleting IL-2 (a T-cell growth factor) and tryptophan; and by destroying target cells via granzyme-mediated cytotoxicity.

Each of the molecules mediating activity of Tregs is of interest for developing ways to control their functions

in experimental setting as well as in clinical practice treating different human diseases.

In many cases, indirect suppressive activity of Tregs is mediated by membrane-bound molecules CTLA-4, LAG-3, and NRP-1 that delay maturation of dendritic cells. Immature dendritic cells have a tolerogenic effect, inferring that certain cytokines they produce during antigen presentation skew differentiation of antigen-specific naive T-cells towards newly generated Treg population, which, in turn, suppress response to some antigenic epitopes. Also, Tregs can influence dendritic cells, resulting in sharply inhibited presentation of the processed antigenic epitopes to helper and cytotoxic T-cells as well as their costimulation. This suppressive mechanism is highly effective, for it is directed against the major players of cellular and humoral adaptive immune response such as Th1-, Th2-, Th17-, follicular helper, and cytotoxic T-cells. Suppression of humoral immune response is related to suppressed interactions between helper T- and B-cells, which is necessary for switching towards synthesis of high-affinity immunoglobulin of IgG, IgA, and IgE classes.

There is no doubt that the mechanism of suppressive action by Tregs is multifaceted and complex and remains not fully clarified. During infection, immunosuppressive mechanisms probably vary depending on the strategy of a particular pathogen, degree of inflammation, etc. (see below).

#### GENERATION OF REGULATORY T-CELLS MEDIATED BY PATHOGENS ACTING ON DENDRITIC CELLS

Differentiation of Tregs (and other T-cell types) requires their interactions with antigen-presenting cells, including dendritic cells. Pathogens co-opt this mechanism for their own purposes. The data presented in this review regarding mechanisms resulting in induction of Tregs mediating immunosuppression triggered by pathogens and their products are summarized in the table. The data presented in this section focuses on examples of Treg generation influenced by dendritic cells mediated by whole-cell pathogens and/or their products. In particular, it was found that bone marrow dendritic cells treated with live *H. pylori* were able to induce Tregs *in vitro* [33]. During experimental infection of mice with *H. pylori*, they were found to generate tolerogenic dendritic cells, which might trigger expansion of Tregs (Foxp3<sup>+</sup> T-cells) and suppress production of proinflammatory cytokines such as IL-17 after adoptive transfer to recipient animals [34]. Perhaps *H. pylori* uses this method to escape from immune clearance (table).

Also, during murine infection caused by Friend's virus (FV), a substantial portion of myeloid dendritic cells was infected with the virus, which impaired their maturation.

## Generation of regulatory T-cells mediated by pathogens and their products

| Inducers                                                                                                                                                                                                                                                    | Target cells                                               | Mechanism of immunosuppression                                                                                                                                                                                                                                                                                                          | References   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1                                                                                                                                                                                                                                                           | 2                                                          | 3                                                                                                                                                                                                                                                                                                                                       | 4            |
| <i>Helicobacter pylori</i> ;<br>Friend's virus; Japanese encephalitis virus                                                                                                                                                                                 | DC                                                         | disturbed DC maturation results in converting naive T-cells into Foxp3 <sup>+</sup> Treg suppressing IL-17 production                                                                                                                                                                                                                   | [33-35]      |
| <i>Bordetella pertussis</i> ,<br>filamentous hemagglutinin adhesin (FHA)                                                                                                                                                                                    | DC                                                         | DCs produce IL-10 followed by differentiation of Tr1 cells suppressing cellular Th1-response (IFN $\gamma$ ); interaction of FHA with integrin $\alpha$ v $\beta$ 3 and $\beta$ 2 C3R on DCs induces development of tolerogenic DCs secreting IL-10 or elimination of effector T-cells                                                  | [36, 52]     |
| <i>Bordetella pertussis</i> ,<br>filamentous hemagglutinin adhesin (FHA)                                                                                                                                                                                    | naive T-cells specific to antigen from <i>B. pertussis</i> | simultaneous activation of antigen-specific TCR and TLR4 induces production of IL-10 by Tregs followed by suppressing inflammation                                                                                                                                                                                                      | [65]         |
| Protein from flagella of motile bacteria, flagellin                                                                                                                                                                                                         | T-cells                                                    | interaction with TLR5 enhances immunosuppressive potential of Foxp3 <sup>+</sup> Tregs                                                                                                                                                                                                                                                  | [64]         |
| Gram-negative bacteria, LPS                                                                                                                                                                                                                                 | CD4 <sup>+</sup> CD25 <sup>+</sup> T-cells                 | interaction of LPS with TLR4 results in sharply upregulated immunosuppressive capacity of antigen-specific Tregs                                                                                                                                                                                                                        | [66]         |
| <i>Schistosoma mansoni</i> :<br>1) lyso-PS lipid<br>2) soluble egg-derived glycoprotein ( $\omega$ -1)                                                                                                                                                      | T-cells                                                    | 1) induced development of IL-10-producing Tr1 cells<br>2) IL-4-producing Foxp3 <sup>+</sup> T-cells are induced that results in suppressed Th1-response                                                                                                                                                                                 | [38]<br>[45] |
| <i>Schistosoma japonicum</i>                                                                                                                                                                                                                                | DC                                                         | upregulated production of IL-10, increased number of CD4 <sup>+</sup> CD25 <sup>+</sup> Foxp3 <sup>+</sup> and CD4 <sup>+</sup> CD25 <sup>+</sup> IL-10 <sup>+</sup> T-cells, downregulated Th1-response, inhibited production of IL-4 and IL-5 by CD4 <sup>+</sup> T-cells, lowered level of eotaxin, suppressed allergic inflammation | [46]         |
| Nematode <i>Ancylostoma caninum</i> , tissue metalloproteinase inhibitor, Ac-TMP-1                                                                                                                                                                          | DC                                                         | generation of CD4 <sup>+</sup> CD25 <sup>+</sup> Foxp3 <sup>+</sup> T-cells producing IL-10 and TGF- $\beta$ , generation of IL-10-producing CD8 <sup>+</sup> CD25 <sup>+</sup> Foxp3 <sup>+</sup> T-cells                                                                                                                              | [39]         |
| Nematode <i>Anisakis simplex</i> , homolog of MIF (macrophage inhibitory factor)                                                                                                                                                                            | APCs, T-cells                                              | upregulated number of Foxp3 <sup>+</sup> T-cells and production of IL-10 and TGF- $\beta$ , suppressed allergic inflammation                                                                                                                                                                                                            | [47]         |
| Nematode <i>Heligmosomoides polygyrus</i> , secretable products                                                                                                                                                                                             | CD11c <sup>lo</sup> CD103 <sup>-</sup> DCs                 | differentiation of Tregs: CD4 <sup>+</sup> CD25 <sup>+</sup> IL-10 <sup>+</sup> Foxp3 <sup>+</sup> and CD4 <sup>+</sup> CD25 <sup>+</sup> IL-10 <sup>+</sup> Foxp3 <sup>-</sup> T-cells                                                                                                                                                 | [40, 41]     |
| <i>Mycobacterium tuberculosis</i> , mannosylated lipoarabinomannan (ManLAM)                                                                                                                                                                                 | monocytes                                                  | induced development of Foxp3 <sup>+</sup> Tregs from CD25 <sup>-</sup> T-cells                                                                                                                                                                                                                                                          | [42]         |
| Persistent pathogens <i>Mycobacterium tuberculosis</i> , <i>Mycobacterium leprae</i> , <i>Helicobacter pylori</i> , <i>Bordetella pertussis</i> , <i>Yersinia pestis</i> , <i>Borrelia burgdorferi</i> , <i>Candida albicans</i> , Morbillivirus, HIV, etc. | DC                                                         | disturbed DC maturation, induced differentiation of Tr1 T-cells, production of IL-10, suppressed cellular Th1- and humoral Th2-response                                                                                                                                                                                                 | [49-51]      |

Table (Contd.)

| 1                                                                                                 | 2                                        | 3                                                                                                                                      | 4        |
|---------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------|
| Bacterial pathogens, LPS and CpG                                                                  | various cell types                       | direct TLR-dependent activated production of IL-10                                                                                     | [54-57]  |
| <i>Yersinia pestis</i> , LcrV protein                                                             | DC                                       | induced production of IL-10, production of IL-12 suppressed via interaction with TLR2/TLR6 and CD1, contributes to Tr1-differentiation | [53, 59] |
| <i>Yersinia enterocolitica</i> , V-antigen                                                        | T-cells                                  | activated TLR2 and CD14 induce production of IL-10 by T-cells                                                                          | [67]     |
| Bacterial superantigens, staphylococcal enterotoxin A and streptococcal pyrogenic toxin A and K/L | T-cells, DC                              | expression of Foxp3 and surface CTLA-4 and CD127 on IL-10-producing T-cells                                                            | [62]     |
| Synthetic bacterial lipoprotein Pam3Cys-SK4                                                       | CD4 <sup>+</sup> CD25 <sup>+</sup> Tregs | interaction of Pam3Cys-SK4 with TLR2 results in upregulated number of CD4 <sup>+</sup> CD25 <sup>+</sup> Tregs                         | [63]     |

Note: DC, dendritic cells; APC, antigen-presenting cell; Treg, regulatory T-cells; Tr1, type 1 regulatory T-cells; TLR, Toll-like receptor; HIV, human immunodeficiency virus.

tion so that when immature dendritic cells contacted naive T-cells, they were skewed towards Treg (Foxp3<sup>+</sup>) differentiation [35].

Differentiation of naive T-cells towards Tregs can be skewed not only by whole-cell microorganisms, but also by their fragments acting via dendritic cells. Infection caused by *Bordetella pertussis* is associated with severe and often fatal lung disease in young children. Filamentous hemagglutinin (FHA) from *B. pertussis* is one of the virulence factors, which acts on the airway dendritic cells and triggers upregulated production of IL-10, resulting in differentiation of Tr1 cells that suppress protective anti-pathogen cellular Th1 response accompanied by lowered production of IFN $\gamma$  [36]. On the other hand, immune escape is provided by a PTx toxin from *B. pertussis*, which can suppress autoimmune reactions through nonspecific stimulation of suppressor T-cells. Thus, this might explain a phenomenon preventing development of autoimmune injury of the central nervous system when mice received PTx toxin at the initial phase of experimental autoimmune encephalomyelitis. This effect was linked to upregulated number of the blood CD4<sup>+</sup>CD25<sup>high</sup>Foxp3<sup>+</sup> Tregs accompanied by abruptly increased serum level of suppressive cytokines IL-10 and TGF- $\beta$  [37].

Lysophosphatidylserine (lyso-PS) from the parasite *Schistosoma mansoni* acting via TLR2 on dendritic cells induced development of Tr1 cells producing IL-10, which contributed not only to escape of the pathogen from the host's immune response, but to dampening severity of immune pathology as well [38].

Likewise, Ac-TMP-1 tissue metalloproteinase inhibitor from the nematode *Ancylostoma caninum* modifies the function of dendritic cells that induce the generation of CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> and CD8<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> T-

cells. Both populations produced IL-10, whereas CD4<sup>+</sup> Tregs additionally synthesized TGF- $\beta$  [39]. In another murine model, it was shown that dendritic cells treated with the products secreted by the nematode *Heligmosomoides polygyrus* preferentially skewed differentiation of naive T-cells towards functionally activated Tregs having the phenotype of CD4<sup>+</sup>CD25<sup>+</sup>IL-10<sup>+</sup>Foxp3<sup>+</sup> T-cells, whereas expansion of the CD11c<sup>lo</sup>CD103<sup>+</sup> dendritic cell subset and Foxp3<sup>+</sup> Tregs was observed in the mesenteric lymph nodes [40, 41]. Apparently, by doing this the helminth tries to establish tolerance towards its antigens, which is necessary for its long-term survival in the host's body.

Experimentally, it was demonstrated that Tregs could develop upon interacting with dendritic cells infected with Japanese Encephalitis Virus (JEV). This might result from upregulated expression of PD-L1 on dendritic cells, which mediates a programmed cell death. PD-1 receptors and their ligands are negative regulators of immune response that belong to the immunoglobulin superfamily and structurally homologous to B7 costimulatory molecules. JEV escapes from the host's immune system by modulating crosstalk between PD-L1 on dendritic cells and PD-1 on T-cells during antigen presentation, thus blocking a TCR (T-cell receptor)-dependent signaling pathway [21].

Infectious agents can act on monocytes and induce development of Foxp3<sup>+</sup> Tregs from CD25<sup>+</sup> T-cells, in particular in response to mannosylated lipoarabinomannan from *Mycobacterium tuberculosis* [42].

It should be emphasized that normal microbiota in the gut avoids hyperinflammatory reactions by using the aforementioned mechanisms to generate tolerogenic dendritic cells, which induce immunosuppression. The regu-

latory activity of Tregs also abolishes immunopathology in the gut, thus having homeostatic potential in the system of the host's body interacting with microbiota [43, 44].

Likewise, autoimmune and allergic processes are also controlled by regulatory T-cells. This explains why pathogen-mediated stimulation of Tregs in experimental settings is manifested not only as suppressed anti-infectious antigen-specific response, but as simultaneously suppressed autoimmune and allergic processes as well. For this viewpoint, the results to be presented in the next section become clear in terms of applied aspects.

Recently, substance  $\omega$ -1 isolated from an egg glycoprotein of *Schistosoma mansoni* was described that prevented development of type 1 diabetes mellitus in NOD mice. It is assumed that this effect was related to the capacity of substance  $\omega$ -1 to induce development of murine Foxp3<sup>+</sup> Tregs and production of IL-4 that together suppress formation of cellular immune response, which has pathogenetic importance during type 1 diabetes mellitus. As a result, destruction of islet  $\beta$ -cells by cytotoxic T-cells was abolished. Importantly, a substance  $\omega$ -1-dependent differentiation of Foxp3<sup>+</sup> T-cells was reproduced in an *in vitro* setting, which also depended on TGF- $\beta$  and retinoic acid. By contrast, IPSE/ $\alpha$ -1, another soluble glycoprotein derived from live eggs of *S. mansoni*, caused no induction of Foxp3<sup>+</sup> Tregs [45].

In another experiment, it was shown that adoptive transfer of dendritic cells isolated from mice infected with *Schistosoma japonicum* (SJ) suppressed allergic reactions in the recipient animals, as dendritic cells from the infected mice produced significant amounts of IL-10 compared to dendritic cells from intact animals. In addition, adoptively transferred dendritic cells resulted in increased count of CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> and CD4<sup>+</sup>CD25<sup>+</sup>IL-10<sup>+</sup> T-cells in recipients, which regulated T-cell response *in vivo* that correlated with suppressed production of IL-4 and IL-5 by CD4<sup>+</sup> T-cells, lowered amount of eotaxin, a chemokine maintaining eosinophilic inflammation in the lungs and, importantly, the degree of suppression for allergic inflammation in recipients [46]. These data suggest that helminth infection can induce tolerogenic dendritic cells that are able to nonspecifically inhibit development of allergic inflammation via upregulating suppressive activity of Tregs.

It was found that the nematode *Anisakis simplex* parasitizing marine animals releases a homolog of MIF (macrophage inhibitory factor) that upregulates the number of Foxp3<sup>+</sup> T-cells after its administration to mice with allergic airway inflammation, and this ameliorates allergic symptoms [47].

Thus, to escape from immune response, pathogens are able to use naturally existing mechanisms responsible for induction of immunosuppression by directly acting on dendritic cells and affecting their maturation as well as suppressing antigen presentation and costimulation of T-cells. Herein, generation of regulatory T-cells was influ-

enced by immature tolerogenic dendritic cells secreting TGF- $\beta$  and IL-10. Virus-induced upregulated expression of PD-L1 on dendritic cells interacting with naive T-cells results in differentiation of regulatory T-cells. Therefore, a pathogen-related tolerogenic activity that lowers the efficacy of cellular and humoral adaptive immune response is mediated by dendritic cells, which influence differentiation of various types of suppressor T-cell subsets able to inhibit anti-pathogen as well as autoimmune and allergic processes.

Tolerogenic dendritic cells develop in response to not only pathogens and their products, but also contacting the normal gut microbiota that prevents potentially hazardous immune reaction against microbiota. It is evident that such interaction between immune cells and microorganisms has features similar to evasion mechanisms, which, on one hand, let them escape from elimination, whereas on the other hand it prevents a hyperinflammatory process.

#### PATHOGEN-INDUCED PRODUCTION OF IMMUNOSUPPRESSIVE CYTOKINE IL-10

It was mentioned above that IL-10 possessing powerful antiinflammatory and immunosuppressive activities is one of the crucial factors controlling immune response (table). This cytokine acts on antigen-presenting cells – T-lymphocytes and other cell types. It is known as a “macrophage-deactivating factor” because it inhibits production of proinflammatory cytokines as well as toxic to microorganisms ROS and RNS in macrophages. IL-10 lowers expression of the MHC class II proteins and costimulatory molecules playing a very important role in antigen presentation and inhibits proliferation of T-cells. It should be noted that IL-10 contributes to differentiation of naive T-cells into antigen-specific regulatory T-cells (Tr1), which, in turn, secrete large amounts of IL-10 as well [13, 48].

A role for IL-10 as an immunoregulatory cytokine was primarily demonstrated during chronic infections. IL-10 downregulates Th1 and Th2 immune response against various pathogens both in experimental settings and in humans upon tuberculosis, hepatitis C, and herpesvirus infection [49]. The majority of immune cells are able to secrete IL-10; however, to complete immune response, it is usually produced at later stages of infection, when the pathogen has been cleared from the body.

Persistent pathogens can induce Tr1-differentiation and IL-10 production by acting on antigen-presenting cells that blocks their maturation. This mechanism is applied by many human pathogens causing clinically relevant diseases, such as *M. tuberculosis*, *M. leprae*, *Helicobacter pylori*, *B. pertussis*, *Yersinia pestis*, *Borrelia burgdorferi*, *Candida albicans*, measles virus, human immunodeficiency virus (HIV), etc. [50].

It was shown that the early-onset and appropriate cellular Th1-response was lowered in animals infected with hypervirulent *Mycobacterium tuberculosis* strain, which correlated with rapid appearance of IL-10-producing Foxp3<sup>+</sup> Tregs [51].

Integrins  $\alpha_v\beta_3$  and  $\beta_2$  (C3R) bound to complement C3 contributed to uptake of apoptotic cells and induced tolerogenic IL-10-secreting dendritic cells, which prevented activation of the immune system by self-antigens contained in apoptotic cells. Similarly, filamentous hemagglutinin adhesin (FHA) from *B. pertussis* binds to C3R and induces immunosuppression [52].

By releasing specific molecules, bacterial pathogens can directly activate production of IL-10, which binds to Toll-like receptors and acts on the host's cell signaling pathways. *IL-10* gene expression can be triggered by TLR ligands such as LcrV, LPS, and CpG that depend on the molecules involved in signaling pathways [53–57].

IL-10-mediated immunosuppression is co-opted by *Yersinia pestis* causing plague to suppress inflammatory response countering replication of the pathogen. Plague modeled in animals conclusively demonstrated that infection with *Y. pestis* downregulated secretion of proinflammatory cytokines such as IFN- $\gamma$  and TNF- $\alpha$ . Mice with *IL-10* gene knockout were relatively resistant to infection with *Y. pestis*, as they developed a normal inflammatory response that resulted in clearance of the pathogen [58]. LcrV protein from *Y. pestis* was shown to induce IL-10 production *in vitro* and suppress secretion of IL-12 by acting via TLR2/TLR6 and CD14 on dendritic cells. It was demonstrated that mice deficient in TLR6 and CD14 were relatively resistant to infection with *Y. pestis*, which reveals a mechanism of immune suppression triggered via TLR2/6 by LcrV as an important virulence factor. It was found that bacterial lipoproteins could activate a TLR2/6-mediated signaling pathway resulting in production of IL-10 by dendritic cells as well as differentiation of regulatory Tr1 cells [53, 59].

IL-27 is a recently identified regulatory factor controlling production of IL-10. IL-27 can limit Th1-, Th2-, and Th17-immune response in various models of infections and autoimmune diseases. Tregs influence dendritic cell production of TGF- $\beta$  and IL-27, which, in turn, induces production of IL-10 by T-cells. This effect depends on transcription factor STAT1 and STAT3 triggered by IL-27 [60].

An effective strategy co-opted by viruses to suppress the response of immune cells is accounted for by production in infected cells of viral homologs of cytokines, chemokines, and their receptors, which antagonize biological effects of the host's interleukins. For instance, HCMV and EBV encode IL-10 viral homologs that inhibit activity of NK cells as well as production of proinflammatory cytokines [1, 11, 61].

It seems that to prevent extensive suppression of effector defense functions, IL-10 must not be continu-

ously produced *in vivo* and is subject to certain regulatory mechanisms acting, particularly, via IL-27.

## PATHOGENS DIRECTLY INFLUENCE DEVELOPMENT OF REGULATORY T-CELLS

There are many examples showing that products of pathogens can directly induce and/or activate Tregs, thus proving the principle that *in vivo* expansion of Tregs is not just a homeostatic response limiting the action of a microbial trigger, but a phenomenon demonstrating that pathogens “learned” to use the ways for augmentation of the host's Tregs for suppressing protective anti-pathogen response.

**Superantigens.** To evade immune response, many pathogenic bacteria and viruses act by using superantigens that disturb immune regulation.

During regular antigen presentation, “MHC + antigenic peptide” complexes are formed, and these interact with antigen-specific naive T-cells. Notably, each antigenic epitope can be contacted only by 1/100,000 lymphocytes. Superantigens may nonspecifically “ligate” MHC class II molecules and TCRs resulting in activation of up to 20% of T-cells, followed by sharply increased production of TNF, IL-1, IL-6, IL-8, IL-10, etc., which are not controlled in their composition and concentration. This results in a “cytokine storm” that gives rise to systemic hyperinflammation and toxic shock. At the same time, regulatory T-cells undergo differentiation, thus further inducing immunosuppression [10].

In particular, staphylococcal enterotoxin A and streptococcal pyrogenic toxin A and K/L caused expression of Foxp3 as well as surface molecules CTLA-4 and CD127 on T-cells. These T-cells possessed high suppressive activity comparable to that of nTregs. Notably, low-dose superantigen-induced CD25<sup>+</sup>Foxp3<sup>+</sup> T-cells produced suppressive cytokine IL-10, but not protective cytokine IFN- $\gamma$  [62].

**TLR signaling pathways.** Infectious pathogens and their components can directly influence Tregs and regulate their activity via receptors of innate immunity known as TLRs. All existing TLRs have been found on their surface and in cytoplasm of T-cells, and expression of TLR4, TLR5, TLR7, and TLR8 was upregulated in Tregs [63].

TLR5 is bound by flagellin, a protein located in flagella in motile microbes that is able to regulate immune response in mucosal sites. Flagellin-mediated costimulation enhanced the suppressive potential of Tregs, which depended on Foxp3 expression [64]. Filamentous hemagglutinin found in flagella of *B. pertussis* can bind to TLR4 on naive antigen-specific T-cells, which converts them into Tregs [65].

It was found that lipopolysaccharide (LPS) acting via TLR4 upregulated expression of the activation markers CD4<sup>+</sup> and CD25<sup>+</sup> on T-cells as well as enhancing the

survival and proliferative activity of the cells. Importantly, their proliferation did not require interaction with antigen-presenting cells, TCR-dependent activation, and IL-2-mediated stimulation. The most crucial finding was that immunosuppressive activity of Tregs was increased by 10-fold in direct response to LPS [66]. Therefore, it represents a case of nonspecific stimulation of Tregs.

In another study, it was found that the TLR2-mediated signaling pathway transiently lowered Foxp3 expression and abolished suppressive phenotype of Tregs [64]. However, other studies demonstrated the opposite — TLR2 agonist stimulated Tregs. In particular, wild-type mice injected with synthetic bacterial lipoprotein Pam3Cys-SK4 as a TLR2 ligand showed significant increase in CD4<sup>+</sup>CD25<sup>+</sup> Tregs [63]. Also, V-antigen from *Yersinia enterocolitica* acted via TLR2 and upregulated IL-10 production that resulted in immunosuppression [67].

Infections caused by the helminth *Schistosoma mansoni* in mice with TLR2 knockout lacked expansion of CD4<sup>+</sup>CD25<sup>+</sup> Tregs, and they developed severe liver pathology. However, adoptive transfer of pathogen-specific CD4<sup>+</sup>CD25<sup>+</sup> Tregs isolated from wild-type animals immunized with *S. mansoni* protected TLR2<sup>-/-</sup> mice from death [68]. In addition, TLR2-deficient mice did not exhibit increased number of Tregs after applying SJMHE1 peptide obtained from heat shock protein HSP-60 of *Schistosoma japonicum* [69]. TLR2 upregulates the activity of Tregs during various infections, particularly demonstrated by relatively enhanced resistance of TLR2<sup>-/-</sup> mice infected by *C. albicans*. TLR2<sup>+</sup> Tregs transferred to TLR2<sup>-/-</sup> mice 100-fold increased *C. albicans* infection, which was prevented in the presence of the TLR2-ligand Pam-3-Cys [63]. Importantly, later it was demonstrated that stimulation of TLR2 on Tregs can directly enhance their proliferation independent of antigen-presenting cells [70], and the suppressive activity of Tregs during this period was reversibly lost [71, 72]. Moreover, other TLR2 ligands including “alarm” signals such as heat shock protein can upregulate the function of Tregs [73]. The controversial role of TLR2 can be explained by the fact that it heterodimerizes with other TLRs (TLR1, TLR6, TLR10). On the other hand, some TLR2 ligands from pathogens can selectively stimulate immunosuppressive, rather than activating, signals [53].

Thus, the question about stimulating or inhibitory action of TLR2 ligands on regulatory T-cells remains unresolved.

Interestingly, TLR8 ligands such as CpG-A and other oligonucleotides can abort immunosuppressive function of CD4<sup>+</sup>CD25<sup>+</sup> Tregs. Short poly-G oligonucleotides G2-G4 vs. G5, G7, and G10 possessed the highest relative suppressive capacity. The mechanism abolishing immunosuppression involved MyD88 adaptor molecule, as the effect disappeared in mice deficient for this molecule. Moreover, suppression of TLR8-mediated

functional activity was noted only in CD4<sup>+</sup>CD25<sup>+</sup> Tregs, but not in effector CD4<sup>+</sup> T-cells [74].

CpG-rich oligodeoxynucleotides as TLR9 ligands can induce proliferation of both CD4<sup>+</sup>CD25<sup>+</sup> and CD4<sup>+</sup>CD25<sup>-</sup> Tregs, which, however, is accompanied by partially abolished suppressive effect of CD4<sup>+</sup>CD25<sup>+</sup> Tregs on effector cells. This was due to a direct stimulatory effect of TLR9 ligand simultaneously acting on effector T-cells, which results in their proliferation and makes them resistant to suppressive effect from Tregs [75].

Based on the abovementioned results, it is evident that TLR ligands can directly stimulate or inhibit Tregs. In particular, ligands for TLR4 and TLR5 upregulate the suppressive effect of Tregs on effector T-cells, whereas ligands for TLR8 and TLR9 abolish or dampen it. Unequivocal results regarding the influence of TLR2-mediated activation of Tregs were obtained that might be due to the experimental setting. Mechanisms underlying the capacity of TLRs to modulate immunosuppressive potential of Tregs remain unclear. On one hand, upregulated or downregulated suppressive activity of Tregs might be related to Foxp3 expression resulting from stimulation of different TLRs that alters the functional activity of CD4<sup>+</sup>CD25<sup>+</sup> Tregs. However, further studies examining the impact of TLR-dependent signaling pathways on Foxp3 expression are needed.

The mechanisms abolishing immunosuppressive effect of Tregs with simultaneous upregulation of proliferative capacity of CD4<sup>+</sup>CD25<sup>+</sup> Tregs after TLR-dependent stimulation might be as follows. It seems that extremely strong activating signals triggering proliferation can eventually result in anergy and apoptosis and abort suppressive activity of CD4<sup>+</sup>CD25<sup>+</sup> Tregs. The data presented here demonstrate that the intensity of immunosuppression can be regulated by pathogens, so that some TLR ligands can upregulate suppressive capacity of Tregs, whereas others may restrain it.

**Pathogen-mediated modification of signaling pathways coupled with receptors of innate immunity.** A mechanism that might be defined as “management of signaling pathways” is among the main strategies used by pathogens, which contributes to development of persistent infections. In particular, they include: pathogen-dependent via host inhibitory receptors, which in some cases can be due to expression of molecules mimicking cognate host ligands [76]; activation of synergetic signaling pathways resulting in production of suppressive mediators such as IL-10 [33] and cAMP [77]; induction of signals resulting in activation of mechanisms underlying safe uptake of pathogens normally used to capture apoptotic bodies [78]; selectively downregulated Th1-response via regulatory complement–TLR interactions [79]; simultaneous triggering of cross-inhibition by different TLR ligands [80] as well as blocking receptors necessary for cooperative induction of defense signals [81]; impaired intracellular signaling pathways due to inactivation of signaling molecules, etc. [82].

Using these and various other mechanisms, pathogens deliberately disintegrate the network of signaling pathways between receptor interactions in the host cells, which results in disrupted regulation of innate immunity.

Typical members of the TLR family have broad but discrete ligand specificity. At the same time, they are barely able to form functional multi-receptor complexes providing an opportunity to recognize multiple pathogenic ligands. Cooperation of ligands results in modification of their pattern-recognition signaling capacity, allowing a host to react to virtually any type of infection, i.e. discriminate among different pathogens, and trigger proper immune response [83].

Immune cells receiving a variety of signals process information, which is subsequently translated through intracellular signaling cascades via transcription factors targeting different genes. By analyzing interactions between intracellular signaling pathways, it was demonstrated that many molecular cascades converge at a limited number of mechanisms of molecular interactions that can have synergistic and antagonistic behavior. Synergy of signaling pathways significantly increases sensitivity of ligand detection due to the fact that united signals triggered by several weak stimuli can cause a strong immune response. Antagonistic interactions between signaling pathways result in upregulating specificity of the response as well as restraining its magnitude, thus preventing collateral tissue damage. Therefore, the interaction of signaling pathways is necessary for normal functioning of the immune system. This allows synergistic activation of the body's defense system to fight against infection, whereas antagonistic interactions prevent potential adverse reactions. Coordinated interactions between signaling pathways serve to maintain a balance between protective immunity and pathological inflammation [84, 85].

Pathogens can induce antagonistic signaling pathways and cause immune suppression, whereas induction of synergistic interaction allows direction of immune reactions along inefficient paths disadvantageous for the host. For instance, *Mycobacterium leprae* triggers humoral Th2-response, whereas protective immunity requires cellular Th1-response [10].

Different pathogens use specific virulence factors to control mechanisms of cooperation between pattern-recognition receptors acting at the level of receptors as well as relevant signaling pathways.

Pathogens co-opt the host's inhibitory receptors to create signaling pathways normally involved in homeostatic interactions that lead to production of immunosuppressive molecules such as IL-10 or cAMP. Pathogens express molecules mimicking relevant ligands that bind to inhibitory receptors [76, 83].

Earlier, it was suggested that IL-10-dependent signaling cascades play an important regulatory role in maintaining homeostasis in the immune system. So,

microorganisms bear genetically determined mechanisms to regulate production of IL-10, and some pathogens can trigger production of IL-10 by inducing synergistic signaling pathways [86]. For example, a number of human pathogens including *Mycobacterium tuberculosis*, *Mycobacterium leprae*, *Helicobacter pylori*, *Candida albicans*, measles virus, and HIV-1 trigger simultaneous activation of TLR and inhibitory receptor DC-SIGN, resulting in conditions when dendritic cells begin to produce large amounts of IL-10 [33]. In particular, binding of mannosylated lipoarabinomannan from *Mycobacterium* spp. (ManLAM) to DC-SIGN initiates a complex signaling cascade that finally activates RAF1 serine/threonine kinase, which, in turn, induces phosphorylation of the p65 subunit of NF- $\kappa$ B transcription factor at Ser276 and subsequent acetylation on several arginine residues. Acetylation of p65 requires signals incoming from both receptors, DC-SIGN and TLR, being mediated by related acetyltransferases such as CREB-binding protein and p300, which are recruited to p65 via binding to phosphorylated Ser276. Acetylation upregulates avidity of NF- $\kappa$ B to DNA and its transcriptional activity, and as well prolongs its presence inside the nucleus so that acetylated NF- $\kappa$ B is able to provide long-lasting upregulated transcription of IL-10. However, at the same time these signaling pathways also induced enhanced transcription of *IL-12* genes potentially able to stimulate T-cell-mediated immune response [87].

The interaction of *H. pylori* with DC-SIGN, mediated by fucose-containing lipopolysaccharide Lewis antigens, causes upregulated production of IL-10 and lowered production of IL-12, which eventually results in downregulating development of Th1 T-cells [88]. Thus, there are different signaling pathways from DC-SIGN that trigger production of IL-10. The spirochete *Borrelia burgdorferi* (the causative agent of Lyme disease) transmitted by the tick *Ixodes scapularis* can induce coactivation of DC-SIGN and TLR2, when its own Salp15 protein is used as a ligand for DC-SIGN. Thus, TLR2 and DC-SIGN are activated simultaneously by spirochetal lipoproteins and tick salivary Salp15 protein pre-absorbed on the surface of *B. burgdorferi*. As a result, TLR-dependent maturation of dendritic cells occurs, which lowers their capacity to activate T-cells and is beneficial for the pathogen as it dampens immune response against these pathogens [89].

Mycobacteria interact with neutrophils using another C-type lectin (CLEC5A) that interacts with ITAM-containing DAP12 adaptor protein and induces an SYK-dependent signaling cascade intercrossing with the TLR2/MYD88-signaling pathway. Synergistic interaction between these pathways activates production of IL-10 due to prolonged phosphorylation of Akt kinase and p38 mitogen-activated protein kinase (MAPK). A murine model of chronic infection demonstrated that IL-10 produced by neutrophils downregulated lung inflammation

that, consequently, accounted for the bacterial load being maintained at a high level [33].

Thus, to escape from immune response, pathogens use different mechanisms of cross activation of inhibitory and activating receptors of innate immunity directed at distorting intracellular signaling pathways.

In physiological settings, a relatively stable state of immune cells is accounted for by regulatory mechanisms that are based on interaction of five components: 1) pathogen that initiates innate defense reactions and regulates immune processes; 2) dendritic cells, which trigger optimal adaptive immune response; 3) helper T-cells inducing and continuously maintaining effector reactions; 4) effector cells directly acting on the pathogen for its eradication; 5) suppressor cells (regulator cells) restricting excessive activity of virtually all immune cells.

Immunosuppressive potential is mediated through various subsets of regulatory CD4<sup>+</sup> T-cells: natural nCD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> and induced iCD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> T-cells. In particular, CD4<sup>+</sup>Foxp3<sup>-</sup> T-cells synthesizing immunosuppressive cytokines IL-10, IL-35 and TGF- $\beta$  along with IFN- $\gamma$  have been described. It should be emphasized that virtually all populations of cells involved in immune response possess suppressive activity by producing relevant cytokines, e.g. B-cells having suppressive potential with respect to T-cells due to IL-10 and TGF- $\beta$ , etc.

Subsets of Tregs differ in terms of their origin, pathways of activation, surface phenotype, as well as mechanisms of suppressive action on target cells. During development of immune response, its magnitude is restrained by additionally induced subsets of Tregs such as Tr1, Th3, and Tr35 T-cells. Tregs target virtually all cells involved in immune response. The influence of Tregs is exhibited as downregulation of immune reactions through production of antiinflammatory cytokines, lowering antigen-presenting function in dendritic cells and macrophages with correspondingly decreased counts of helper CD4<sup>+</sup> Th1, Th2, Th17, and cytotoxic CD8<sup>+</sup> T-cells as well as cytokines produced by them, and induction of apoptosis. Immunosuppression is entirely pointed at ameliorating damaging effect of local and systemic hyperinflammatory reactions accompanying immune response. Thus, immunosuppression in some cases reduces efficacy of immune response, facilitating long-term persistence of pathogens *in vivo*. Apparently, successful eradication of pathogen is accomplished due to a balance between effector and regulatory defense mechanisms, and its disturbance underlies unfavorable course of infectious processes.

Various pathogens can induce different preferred suppressive mechanisms so that persistent pathogens are able to survive for decades in the host's body. The efficacy of immune response against such pathogens seems to be insufficient, as pathogens use sophisticated evasion strategies aimed at escape, disturbance, distortion, and suppression of immune defense reactions in the host.

Here, it was demonstrated that manipulation of a network consisting of regulatory T-cells is among the most important strategies used by viruses, bacteria, protozoa, helminths, and fungi to ensure their survival and further spread of their genome. Interestingly, some pathogens are able to directly induce conversion of naive T-cells into suppressive Foxp3<sup>+</sup> Tregs, whereas others activate preexisting natural Tregs: in both cases, this results in suppression of anti-pathogen effector response. Microbe-associated disturbances in maturation of dendritic cells, stimulation through TLRs or other pattern recognition receptors, induced production of cytokines, and release of signaling "danger" molecules by injured tissues – altogether, they contribute to differentiation of induced iTregs, therefore supporting survival of pathogens. One of the tactics used by microorganisms is their simultaneous interaction with activating and inhibitory receptors on the cells of innate immunity. Multiple combinations of signals induced in such way result in development of unusual transduction within intracellular signaling pathways, which, can subsequently lead to synthesis of mediators suppressing immune response.

The capacity to replicate and spread their own genome upon parasitizing in the host's body results from the long-term coevolution between micro- and macroorganisms. Mechanisms of host defense immune reactions have been gradually improved during such coevolution, thus providing the host with a transient advantage. However, it is countered by a more accelerated evolution of pathogens that provides a new impetus for further upgrade of immune mechanisms. Based on this viewpoint, a parasite anticipates the host. However, eventually both parasite and host act as an obligatory source of life for the parasite to reproduce its genome [90].

Evasion programs used by pathogens that were discussed in this review demonstrate possible opportunities for control after Treg activity provided by products of pathogens, which may be needed upon developing curative preparations to correct autoimmune, allergic, and oncologic diseases.

Along with this, it is known that induction of immunosuppression due to certain Treg-stimulating ligands contained in vaccine seriously restrains its efficacy. Apparently, engineering of new vaccines lacking such suppressive effect by using counteracting ligands will provide an optimal immune response after vaccination.

The authors are very grateful to Dr. S. A. Nedospasov and the reviewers for their valuable advice and comments they shared while preparing this review for publication.

## REFERENCES

1. Playfair, J. H. L., and Bancroft, G. J. (2012) *Infection and Immunity*, 4th Edn., Oxford University Press, Oxford, pp. 115-120.

2. Tischler, A. D., and McKinney, J. D. (2011) in *The Immune Response to Infection* (Kaufmann, S. H. E., Rouse, B. T., and Sachs, D. L., eds.) ASM Press, Washington, DC, pp. 425-440.
3. Sansonetti, P. J., and Puhar, A. (2011) in *The Immune Response to Infection* (Kaufmann, S. H. E., Rouse, B. T., and Sachs, D. L., eds.) ASM Press, Washington, DC, pp. 133-142.
4. Garib, F. Yu. (2013) *Interaction of Pathogens with Innate Immunity* [in Russian], MSU, Moscow.
5. Garib, F. Yu., and Rizopulu, A. P. (2012) Interaction of pathogenic bacteria with host innate immunity, *Infekts. Immun.*, **62**, 581-596.
6. Gal-Mor, O., and Finlay, B. B. (2006) Pathogenicity islands: a molecular toolbox for bacterial virulence, *Cell Microbiol.*, **8**, 1707-1719.
7. Forsberg, A., Rosqvist, R., and Fallman, M. (2003) in *Bacterial Evasion in Host Immune Responses* (Henderson, B., and Oyston, P. C. E., eds.) Cambridge University Press, pp. 127-170.
8. Pritchard, D., Hooi, D., Watson, E., Chow, S., Telford, G., Bycroft, B., Chhabra, C., Harty, C., Camara, M., Diggle, S., and Williams, P. (2003) in *Bacterial Evasion in Host Immune Responses* (Henderson, B., and Oyston, P. C. E., eds.) Cambridge University Press, pp. 201-222.
9. Mesman, A. W., Zijlstra-Willems, E. M., Kaptein, T. M., de Swart, R. L., Davis, M. E., Ludlow, M., Duprex, W. P., Gack, M. U., Gringhuis, S. I., and Geijtenbeek, T. B. (2014) Measles virus suppresses RIG-I-like receptor activation in dendritic cells via DC-SIGN-mediated inhibition of PP1 phosphatases, *Cell Host Microbe*, **16**, 31-42.
10. Murphy, K. P. (2012) *Janeway's Immunobiology*, Garland Science, Taylor and Francis group, LLC.
11. Farrington, L., O'Neill, G., and Hill, A. B. (2011) in *The Immune Response to Infection* (Kaufmann, S. H. E., Rouse, B. T., and Sachs, D. L., eds.) ASM Press, Washington, DC, pp. 393-401.
12. Sakaguchi, S., Vignali, D. A., Rudensky, A. Y., Niec, R. E., and Waldmann, H. (2013) The plasticity and stability of regulatory T-cells, *Nat. Rev. Immunol.*, **13**, 461-467.
13. Shevach, E. M. (2013) in *Fundamental Immunology* (Paul, W. E., ed.) Lippincott Williams and Wilkins, pp. 795-832.
14. Buckner, J. H. (2010) Mechanisms of impaired regulation by CD4+CD25+FOXP3+ regulatory T-cells in human autoimmune diseases, *Nat. Rev. Immunol.*, **10**, 849-859.
15. Wang, R., Wan, Q., Kozhaya, L., Fujii, H., and Unutmaz, D. (2008) Identification of a regulatory T-cell specific cell surface molecule that mediates suppressive signals and induces Foxp3 expression, *PLoS One*, **3**, e2705.
16. Collison, L. W., Chaturvedi, V., Henderson, A. L., Giacomini, P. R., Guy, C., Bankoti, J., Finkelstein, D., Forbes, K., Workman, C. J., Brown, S. A., Rehg, J. E., Jones, M. L., Ni, H. T., Artis, D., Turk, M. J., and Vignali, D. A. (2010) IL-35-mediated induction of a potent regulatory T-cell population, *Nat. Immunol.*, **11**, 1093-1101.
17. Jankovic, D., Kullberg, M. C., Feng, C. G., Goldszmid, R. S., Collazo, C. M., Wilson, M., Wynn, T. A., Kamanaka, M., Flavell, R. A., and Sher, A. (2007) Conventional T-bet+Foxp3- Th1-cells are the major source of host-protective regulatory IL-10 during intracellular protozoan infection, *J. Exp. Med.*, **204**, 273-283.
18. Nakagawa, T., Tsuruoka, M., Ogura, H., Okuyama, Y., Arima, Y., Hirano, T., and Murakami, M. (2010) IL-6 positively regulates Foxp3+CD8+ T-cells *in vivo*, *Int. Immunol.*, **22**, 129-139.
19. Rubtsov, Y. P., Niec, R. E., Josefowicz, S., Li, L., Darce, J., Mathis, D., Benoist, C., and Rudensky, A. Y. (2010) Stability of the regulatory T-cell lineage *in vivo*, *Science*, **329**, 1667-1671.
20. Shevach, E. M. (2009) Mechanisms of Foxp3+ T-regulatory cell-mediated suppression, *Immunity*, **30**, 636-645.
21. Gupta, N., Hegde, P., Lecerf, M., Nain, M., Kaur, M., Kalia, M., Vratil, S., Bayry, J., Lacroix-Desmazes, S., and Kaveri, S. V. (2014) Japanese encephalitis virus expands regulatory T-cells by increasing the expression of PD-L1 on dendritic cells, *Eur. J. Immunol.*, **44**, 1363-1374.
22. Chaturvedi, V., Collison, L. W., Guy, C. S., Workman, C. J., and Vignali, D. A. (2011) Cutting edge: human regulatory T-cells require IL-35 to mediate suppression and infectious tolerance, *J. Immunol.*, **186**, 6661-6666.
23. Vignali, D. A., Collison, L. W., and Workman, C. J. (2008) How regulatory T-cells work, *Nature Rev. Immunol.*, **8**, 523-532.
24. Cao, X., Cai, S. F., Fehniger, T. A., Song, J., Collins, L. I., Piwnicka, D. R., and Ley, T. J. (2007) Granzyme B and perforin are important for regulatory T-cell-mediated suppression of tumor clearance, *Immunity*, **27**, 635-646.
25. Garin, M. I., Chu, C. C., Golshayan, D., Cernuda-Morollon, E., Wait, R., and Lechler, R. I. (2007) Galectin-1: a key effector of regulation mediated by CD4+CD25+ T-cells, *Blood*, **109**, 2058-2065.
26. Deaglio, S., Dwyer, K. M., Gao, W., Friedman, D., Usheva, A., Erat, A., Chen, J. F., Enjyoji, K., Linden, J., Oukka, M., Kuchroo, V. K., Strom, T. B., and Robson, S. C. (2007) Adenosine generation catalyzed by CD39 and CD73 expressed on regulatory T-cells mediates immune suppression, *J. Exp. Med.*, **204**, 1257-1265.
27. Mandapathil, M., Hilldorfer, B., Szczepanski, M. J., Czystowska, M., Szajnik, M., Ren, J., Lang, S., Jackson, E. K., Gorelik, E., and Whiteside, T. L. (2010) Generation and accumulation of immunosuppressive adenosine by human CD4+CD25highFOXP3+ regulatory T-cells, *J. Biol. Chem.*, **285**, 7176-7186.
28. Lukashev, D., Ohta, A., Apasov, S., Chen, J. F., and Sitkovsky, M. (2004) Cutting edge: physiologic attenuation of proinflammatory transcription by the Gs protein-coupled A2A adenosine receptor *in vivo*, *J. Immunol.*, **173**, 21-24.
29. Thammavongsa, V., Kern, J. W., Missiakas, D. M., and Schneewind, O. (2009) *Staphylococcus aureus* synthesizes adenosine to escape host immune responses, *J. Exp. Med.*, **206**, 2417-2427.
30. Zarek, P. E., Huang, C. T., Lutz, E. R., Kowalski, J., Horton, M. R., Linden, J., Drake, C. G., and Powell, J. D. (2008) A2A receptor signaling promotes peripheral tolerance by inducing T-cell energy and the generation of adaptive regulatory T-cells, *Blood*, **111**, 251-259.
31. Liang, B. T., Workman, C., Lee, J., Chew, C., Dale, B. M., Colonna, L., Flores, M., Li, N. Y., Schweighoffer, E., Greenberg, S., Tybulewicz, V., Vignali, D., and Clynes, R. (2008) Regulatory T-cells inhibit dendritic cells by lymphocyte activation gene-3 engagement of MHC class II, *J. Immunol.*, **180**, 5916-5926.
32. Sarris, M., Anderson, K. G., Randow, F., Mayr, L., and Betz, A. G. (2008) Neuropilin-1 expression on regulatory T-cells enhances their interactions with dendritic cells during antigen recognition, *Immunity*, **28**, 402-413.

33. Zhang, M., Liu, M., Luther, J., and Kao, J. Y. (2010) *Helicobacter pylori* directs tolerogenic programming of dendritic cells, *Gut Microbes*, **1**, 325-329.
34. Kao, J. Y., Zhang, M., Miller, M. J., Mills, J. C., Wang, B., Liu, M., Eaton, K. A., Zou, W., Berndt, B. E., Cole, T. S., Takeuchi, T., Owyang, S. Y., and Luther, J. (2010) *Helicobacter pylori* immune escape is mediated by dendritic cell-induced Treg-skewing and Th17 suppression in mice, *Gastroenterology*, **138**, 1046-1054.
35. Balkow, S., Krux, F., Loser, K., Becker, J. U., Grabbe, S., and Dittmer, U. (2007) Friend retrovirus infection of myeloid dendritic cells impairs maturation, prolongs contact to naive T-cells, and favors expansion of regulatory T-cells, *Blood*, **110**, 3949-3958.
36. McGuirk, P., McCann, C., and Mills, K. H. G. (2002) Pathogen-specific T-regulatory 1 cells induced in the respiratory tract by a bacterial molecule that stimulates interleukin 10 production by dendritic cells: a novel strategy for evasion of protective T-helper type 1 responses by *Bordetella pertussis*, *J. Exp. Med.*, **195**, 221-231.
37. Weber, M. S., Benkhoucha, M., Lehmann-Horn, K., Hertenberg, D., Sellner, J., Santiago-Raber, M. L., Chofflon, M., Hemmer, B., Zamvil, S. S., and Lalive, P. H. (2010) Repetitive pertussis toxin promotes development of regulatory T-cells and prevents central nervous system autoimmune disease, *PLoS One* **5**, e16009.
38. Van der Kleij, D., Latz, E., Brouwers, J. F., Kruize, Y. C., Schmitz, M., Kurt-Jones, E. A., Espevik, T., de Jong, E. C., Kapsenberg, M. L., Golenbock, D. T., Tielens, A. G., and Yazdanbakhsh, M. (2002) A novel host-parasite lipid cross talk: schistosomal lysophosphatidylserine activates Toll-like receptor 2 and affects immune polarization, *J. Biol. Chem.*, **277**, 48122-48129.
39. Cue'llar, C., Wu, W., and Mendez, S. (2009) The hookworm tissue inhibitor of metalloproteases (Ac-TMP-1) modifies dendritic cell function and induces generation of CD4 and CD8 suppressor T-cells, *PLoS Negl. Trop. Dis.*, **3**, e439.
40. Segura, M., Su, Z., Piccirillo, C., and Stevenson, M. M. (2007) Impairment of dendritic cell function by excretory-secretory products: a potential mechanism for nematode-induced immunosuppression, *Eur. J. Immunol.*, **37**, 1887-1904.
41. Smith, K. A., Hochweller, K., Hammerling, G. J., Boon, L., Macdonald, A. S., and Maizels, R. M. (2011) Chronic helminth infection promotes immune regulation *in vivo* through dominance of CD11c<sup>+</sup>CD103<sup>+</sup> dendritic cells, *J. Immunol.*, **186**, 7098-7109.
42. Chieppa, M., Bianchi, G., Doni, A., Del Prete, A., Sironi, M., Laskarin, G., Monti, P., Piemonti, L., Biondi, A., Mantovani, A., Intra, M., and Allavena, P. (2003) Cross-linking of the mannose receptor on monocyte-derived dendritic cells activates an anti-inflammatory immunosuppressive program, *J. Immunol.*, **171**, 4552-4560.
43. Josefowicz, S. Z., Ni, R. E., Kim, H. Y., Treuting, P., Chinen, T., Zheng, Y., Umetsu, D. T., and Rudensky, A. Y. (2012) Extrathymically generated regulatory T-cells control mucosal TH2 inflammation, *Nature*, **482**, 395-399.
44. Tanoue, T., and Honda, K. (2012) Induction of Treg-cells in the mouse colonic mucosa: a central mechanism to maintain host-microbiota homeostasis, *Sem. Immunol.*, **24**, 50-57.
45. Zacccone, P., Burton, O. T., Gibbs, S. E., Miller, N., Jones, F. M., Schramm, G., Haas, H., Doenhoff, M. J., Dunne, D. W., and Cooke, A. (2011) The *S. mansoni* glycoprotein  $\omega$ -1 induces Foxp3 expression in NOD mouse CD4 T-cells, *Eur. J. Immunol.*, **41**, 2709-2718.
46. Liu, J. Y., Li, L. Y., Yang, X. Z., Li, J., Zhong, G., Wang, J., Li, L. J., Ji, B., Wu, Z. Q., Liu, H., Yang, X., and Liu, P. M. (2011) Adoptive transfer of DCs isolated from helminth-infected mice enhanced T-regulatory cell responses in airway allergic inflammation, *Parasite Immunol.*, **33**, 525-534.
47. Park, S. K., Cho, M. K., Park, H. K., Lee, K. H., Lee, S. J., Choi, S. H., Ock, M. S., Jeong, H. J., Lee, M. H., and Yu, H. S. (2009) Macrophage migration inhibitory factor homologs of anisakis simplex suppress Th2 response in allergic airway inflammation model via CD4<sup>+</sup>CD25<sup>+</sup>Foxp3<sup>+</sup> T-cell recruitment, *J. Immunol.*, **182**, 6907-6914.
48. Sakaguchi, S., Wing, K., and Miara, M. (2013) in *Clinical Immunology: Principles and Practice*, Elsevier, pp. 193-202.
49. Sarangi, P. P., Sehrawat, S., Suvas, S., and Rouse, B. T. (2008) IL-10 and natural regulatory T-cells: two independent anti-inflammatory mechanisms in herpes simplex virus-induced ocular immunopathology, *J. Immunol.*, **180**, 6297-6306.
50. Belkaid, Y., and Tarbell, K. (2009) Regulatory T-cells in the control of host-microorganism interactions, *Ann. Rev. Immunol.*, **27**, 551-589.
51. Ordway, D., Henao-Tamayo, M., Harton, M., Palanisamy, G., Trout, J., Shanley, C., Basaraba, R. J., and Orme, I. M. (2007) The hypervirulent *Mycobacterium tuberculosis* strain HN878 induces a potent TH1 response followed by rapid down-regulation, *J. Immunol.*, **179**, 522-531.
52. Mahnke, K., Knop, J., and Enk, A. H. (2003) Induction of tolerogenic DCs: "you are what you eat", *Trends Immunol.*, **24**, 646-651.
53. DePaolo, R. W., Tang, F., Kim, I., Han, M., Levin, N., Ciletti, N., Lin, A., Anderson, D., Schneewind, O., and Jabri, B. (2008) Toll-like receptor 6 drives differentiation of tolerogenic dendritic cells and contributes to LcrV-mediated plague pathogenesis, *Cell Host Microbe*, **4**, 350-361.
54. Medzhitov, R. (2008) in *Fundamental Immunology* (Paul, W. E., ed.) Lippincott Williams and Wilkins, pp. 427-450.
55. Mion, F., Tonon, S., Toffoletto, B., Cesselli, D., Pucillo, C. E., and Vitale, G. (2014) IL-10 production by B-cells is differentially regulated by immune-mediated and infectious stimuli and requires p38 activation, *Mol. Immunol.*, **62**, 266-276.
56. Carey, A. J., Tan, C. K., and Ulett, G. C. (2012) Infection-induced IL-10 and JAK-STAT: a review of the molecular circuitry controlling immune hyperactivity in response to pathogenic microbes, *JAKSTAT*, **1**, 159-167.
57. Ketlinskiy, S. A., and Simbirtsev, A. S. (2008) *Cytokines* [in Russian], Izdatel'stvo Foliant, St. Petersburg.
58. Brubaker, R. R. (2003) Interleukin-10 and inhibition of innate immunity to *Yersinia*: roles of Yops and LcrV (V-antigen), *Infect. Immun.*, **71**, 3673-3681.
59. Kopp, E., and Medzhitov, R. (2002) A plague on host defense, *J. Exp. Med.*, **21**, 1009-1012.
60. Stumhofer, J. S., Silver, J. S., Laurence, A., Porrett, P. M., Harris, T. H., Turka, L. A., Ernst, M., Saris, C. J., O'Shea, J. J., and Hunter, C. A. (2007) Interleukins 27 and 6 induce STAT3-mediated T-cell production of interleukin 10, *Nat. Immunol.*, **8**, 1363-1371.
61. Alcamí, A., and Saraiva, M. (2009) Chemokine binding proteins encoded by pathogens, *Adv. Exp. Med. Biol.*, **666**, 167-179.

62. Taylor, A. L., and Llewelyn, M. J. (2010) Superantigen-induced proliferation of human CD4+CD25- T-cells is followed by a switch to a functional regulatory phenotype, *J. Immunol.*, **185**, 6591-6598.
63. Suttmüller, R. P. M., Morgan, M. E., Netea, M. G., Grauer, O., and Adema, G. J. (2006) Toll-like receptors on regulatory T-cells: expanding immune regulation, *Trends Immunol.*, **27**, 387-393.
64. Crellin, N. K., Garcia, R. V., Hadisfar, O., Allan, S. E., Steiner, T. S., and Levings, M. K. (2005) Human CD4+ T-cells express TLR5 and its ligand flagellin enhances the suppressive capacity and expression of FOXP3 in CD4+CD25+ T regulatory cells, *J. Immunol.*, **175**, 8051-8059.
65. Higgins, S. C., Lavelle, E. C., McCann, C., Keogh, B., McNeela, E., Byrne, P., O'Gorman, B., Jarnicki, A., McGuirk, P., and Mills, K. H. (2003) Toll-like receptor 4-mediated innate IL-10 activates antigen-specific regulatory T-cells and confers resistance to *Bordetella pertussis* by inhibiting inflammatory pathology, *J. Immunol.*, **171**, 3119-3127.
66. Caramalho, I., Lopes-Carvalho, T., Ostler, D., Zelenay, S., Haury, M., and Demengeot, J. (2003) Regulatory T-cells selectively express toll-like receptors and are activated by lipopolysaccharide, *J. Exp. Med.*, **197**, 403-411.
67. Sing, A., Rost, D., Tvardovskaia, N., Roggenkamp, A., Wiedemann, A., Kirschning, C. J., Aepfelbacher, M., and Heesemann, J. (2002) *Yersinia* V-antigen exploits toll-like receptor 2 and CD14 for interleukin 10-mediated immunosuppression, *J. Exp. Med.*, **196**, 1017-1024.
68. Layland, L. E., Rad, R., Wagner, H., and da Costa, C. U. (2007) Immunopathology in schistosomiasis is controlled by antigen-specific regulatory T-cells primed in the presence of TLR2, *Eur. J. Immunol.*, **37**, 2174-2184.
69. Wang, X., Zhou, S., Chi, Y., Wen, X., Hoellwarth, J., He, L., Liu, F., Wu, C., Dhessi, S., Zhao, J., Hu, W., and Su, C. (2009) CD4+CD25+Treg induction by an HSP60-derived peptide SJMHE1 from *Schistosoma japonicum* is TLR2 dependent, *Eur. J. Immunol.*, **39**, 3052-3065.
70. Chen, Q., Davidson, T. S., Huter, E. N., and Shevach, E. M. (2009) Engagement of TLR2 does not reverse the suppressor function of mouse regulatory T-cells, but promotes their survival, *J. Immunol.*, **183**, 4458-4466.
71. Oberg, H. H., Ly, T. T., Ussat, S., Meyer, T., Kabelitz, D., and Wesch, D. (2010) Differential but direct abolishment of human regulatory T-cell suppressive capacity by various TLR2 ligands, *J. Immunol.*, **184**, 4733-4740.
72. Van Maren, W. W., Nierkens, S., Toonen, L. W., Bolscher, J. M., Suttmüller, R. P., and Adema, G. J. (2011) Multifaceted effects of synthetic TLR2 ligand and *Legionella pneumophila* on Treg-mediated suppression of T-cell activation, *BMC Immunol.*, DOI: 10.1186/1471-2172-12-23.
73. Zanin-Zhorov, A., Cahalon, L., Tal, G., Margalit, R., Lider, O., and Cohen, I. R. (2006) Heat shock protein 60 enhances CD4+CD25+ regulatory T-cell function via innate TLR2 signaling, *J. Clin. Invest.*, **116**, 2022-2032.
74. Peng, G., Guo, Z., Kiniwa, Y., Voo, K. S., Peng, W., Fu, T., Wang, D. Y., Li, Y., Wang, H. Y., and Wang, R. F. (2005) Toll-like receptor 8-mediated reversal of CD4+ regulatory T-cell function, *Science*, **309**, 1380-1384.
75. Chiffolleau, E., Heslan, J. M., Heslan, M., Louvet, C., Condamine, T., and Cuturi, M. C. (2007) TLR9 ligand enhances proliferation of rat CD4+ T-cell and modulates suppressive activity mediated by CD4+CD25+ T-cell, *Int. Immunol.*, **19**, 193-201.
76. Carlin, A. F., Uchiyama, S., Chang, Y. C., Lewis, A. L., Nizet, V., and Varki, A. (2009) Molecular mimicry of host sialylated glycans allows a bacterial pathogen to engage neutrophil Siglec-9 and dampen the innate immune response, *Blood*, **113**, 3333-3336.
77. Wang, M., Krauss, J. L., Domon, H., Hosur, K. B., Liang, S., Magotti, P., Triantafyllou, M., Triantafyllou, K., Lambris, J. D., and Hajishengallis, G. (2010) Microbial hijacking of complement-Toll-like receptor crosstalk, *Sci. Signal.*, **3**, DOI: 10.1126.
78. Oliva, C., Turnbough, C. L., Jr., and Kearney, J. F. (2009) CD14-Mac-1 interactions in *Bacillus anthracis* spore internalization by macrophages, *Proc. Natl. Acad. Sci. USA*, **106**, 13957-13962.
79. Hajishengallis, G., and Lambris, J. D. (2010) Crosstalk pathways between Toll-like receptors and the complement system, *Trends Immunol.*, **31**, 154-163.
80. Simmons, D. P., Canaday, D. H., Liu, Y., Li, Q., Huang, A., Boom, W. H., and Harding, C. V. (2010) *Mycobacterium tuberculosis* and TLR2 agonists inhibit induction of type I IFN and class I MHC antigen cross processing by TLR9, *J. Immunol.*, **185**, 2405-2415.
81. Melendez, A. J., Harnett, M. M., Pushparaj, P. N., Wong, W. S., Tay, H. K., McSharry, C. P., and Harnett, W. (2007) Inhibition of FcεRI-mediated mast cell responses by ES-62, a product of parasitic filarial nematodes, *Nat. Med.*, **13**, 1375-1381.
82. Brodsky, I. E., and Medzhitov, R. (2009) Targeting of immune signaling networks by bacterial pathogens, *Nat. Cell Biol.*, **11**, 521-526.
83. Hajishengallis, G., and Lambris, J. D. (2011) Microbial manipulation of receptor crosstalk in innate immunity, *Nat. Rev. Immunol.*, **11**, 187-200.
84. Ivashkiv, L. B. (2009) Cross-regulation of signaling by ITAM associated receptors, *Nat. Immunol.*, **10**, 340-347.
85. Zak, D. E., and Aderem, A. (2009) Systems biology of innate immunity, *Immunol. Rev.*, **227**, 264-282.
86. Gringhuis, S. I., den Dunnen, J., Litjens, M., van Het Hof, B., van Kooyk, Y., and Geijtenbeek, T. B. (2007) C-type lectin DC-SIGN modulates Toll-like receptor signaling via Raf-1 kinase-dependent acetylation of transcription factor NF-κB, *Immunity*, **26**, 605-616.
87. Gringhuis, S. I., den Dunnen, J., Litjens, M., van der Vlist, M., and Geijtenbeek, T. B. (2009) Carbohydrate-specific signaling through the DC-SIGN signalosome tailors immunity to *Mycobacterium tuberculosis*, HIV-1 and *Helicobacter pylori*, *Nat. Immunol.*, **10**, 1081-1088.
88. Bergman, M. P., Engering, A., Smits, H. H., van Vliet, S. J., van Bodegraven, A. A., Wirth, H. P., Kapsenberg, M. L., Vandenbroucke-Grauls, C. M., van Kooyk, Y., and Appelmek, B. J. (2004) *Helicobacter pylori* modulates the T helper cell 1/T helper cell 2 balance through phase-variable interaction between lipopolysaccharide and DC-SIGN, *J. Exp. Med.*, **200**, 979-990.
89. Hovius, J. W., de Jong, M. A., den Dunnen, J., Litjens, M., Fikrig, E., van der Poll, T., Gringhuis, S. I., and Geijtenbeek, T. B. (2008) Salp15 binding to DC-SIGN inhibits cytokine expression by impairing both nucleosome remodeling and mRNA stabilization, *PLoS Pathog.*, **4**, e31.
90. Hedrick, S. M. (2004) The acquired immune system: a vantage from beneath, *Immunity*, **21**, 607-615.